

FIN ST1182–ST1271

Strategic Three-Package Architecture and Operational Completion

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Six research rounds — 90 programmes

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This report updates the earlier FIN strategic verdict. It preserves the strict informational core $W0$, the conditional conversion package CA , and the conditional sector package SA , but proves that they do not yet define a unique operational theory. A third independent package OA is necessary. No conditional object is promoted to the strict core.

Abstract

The earlier strategic verdict proposed a two-package architecture: a strict dimensionless informational core $W0$, conversion axioms CA for units, and sector axioms SA for origin/polarity selection, producing conditional effective physics $W3$. Release 10.96 subsequently proved that the same strict operator A admits exact classical and quantum state models with incompatible dynamics. ST1182–ST1271 test whether $CA + SA$ removes that ambiguity.

First, the strict-source intake is closed without replay: no new scale-charged S_+ or translation-breaking source has appeared. On the finite carrier the sampled strict kernel and A are algebraically interdefinable. The functional calculus algebra $C^*(A)$ is commutative and seven-dimensional. Adding the vertex diagonal algebra generates the full M_{12} because every off-diagonal entry of A is nonzero, but choosing that diagonal algebra as physical input is itself extra structure and still permits both classical and quantum uses.

The conversion package is reconstructed as the positive triple $(\ell_*, \tau_*, \hbar_*)$. It is the unique rank-three subset spanning length, time and action, and yields

$$c_* = \ell_*/\tau_*, \quad E_* = \hbar_*/\tau_*, \quad p_* = \hbar_*/\ell_*, \quad m_* = \hbar_*\tau_*/\ell_*^2.$$

Its numerical values remain a free $(\mathbb{R}_+)^3$ torsor. It does not select a sector, state category, temperature scale or instrument.

The sector package is reconstructed on the frame torsor $\mathbb{Z}_{12} \times \{\pm 1\}$. The dihedral action is free and transitive. Origin plus polarity select one frame, and a coupling axiom makes that frame dynamically relevant, but all are explicitly non-strict. The unique invariant probability measure is uniform and cannot select a frame.

The main theorem falsifies the sufficiency of the earlier two-package architecture. Three pairwise inequivalent models share exactly the same $W0$, CA and SA . For the same selected-vertex preparation at dimensionless time one, classical heat predicts return probability 0.2628294053, whereas unitary evolution predicts 0.5843533247. Thus units and sector choice do not choose state or dynamics.

An independent operational/state package OA is therefore unavoidable. Its minimal interface contains StateCategory, PhysicalChannel, ClockMap, Preparation, Instrument and RecordRule. Removing any one leaves a concrete ambiguity or destroys falsifiability. The strategic architecture must be upgraded to

$$W0 + (CA \times SA \times OA) \longrightarrow W3_{\text{conditional}}.$$

This is a conditional model-building framework, not strict ToE closure.

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1 Strategic input and nonreplay rule

The current repository verdict separates three previously known cuts: dimension, sector selection and the information-to-physics bridge. It also closes repeated searches in invariant/weight-zero receiver classes. Release 10.96 adds a fourth fact: the operator datum does not select classical versus quantum state semantics.

Accordingly, this cycle did not repeat generic legacy-to-strict fitting, invariant selector searches, dimensionless S_+ monomials, phase locking or boundary fitting to c . No new hard source formula entered the repository. The only admissible local continuation was to formalize and falsify the explicit conditional architecture.

2 Round I: ST1182–ST1196 — strict core $W0$ and algebra

The strict informational ledger is

$$W0 = \{K_{\text{strict}}, A, \alpha_{\text{geo}}, A_\phi, \text{finite spectral/graph theorems and no-go certificates}\},$$

where

$$\alpha_{\text{geo}} = 4 \ln 2, qquad A_\phi = \frac{2\pi}{\alpha_{\text{geo}}}, qquad \alpha_{\text{geo}} A_\phi = 2\pi.$$

These identities remain dimensionless and do not define $\hbar$.

On the finite carrier,

$$W = sI - A, qquad A = sI - W, qquad s = 1.660307278766099.$$

Therefore the sampled strict kernel matrix contains no additional state- category selector beyond (A, s) . This statement does not collapse the full continuous kernel function to its twelve-point sample.

The spectrum has seven distinct values, so

$$C^*(A) \cong \mathbb{C}^7$$

is commutative. It cannot by itself be the full noncommutative observable algebra M_{12} .

There is nevertheless a useful algebraic result. Let $\mathcal{D} = \text{span}\{E_{ii}\}$ be the declared vertex diagonal algebra. Since $A_{ij} \neq 0$ for every $i \neq j$,

$$E_{ii} A E_{jj} = A_{ij} E_{ij}.$$

Hence $\mathcal{D}$ and A generate every matrix unit and therefore M_{12} . This proves algebraic reachability, not physical quantum semantics: the same vertex data also defines the classical Markov model.

3 Round II: ST1197–ST1211 — conversion package CA

Define the explicitly conditional positive scales

$$CA = (\ell_*, \tau_*, \hbar_*) \in (\mathbb{R}_+)^3$$

with dimensions length, time and action. The exhaustive audit of all eight subsets gives rank three only for the full triple; every proper subset has rank at most two.

The package consistently dimensionalizes both main channels:

$$H_{\text{phys}} = \frac{\hbar_*}{\tau_*} A, qquad U(t_{\text{phys}}) = \exp\left[-i \frac{t_{\text{phys}}}{\tau_*} A\right],$$

$$L_{\text{heat}} = -\frac{1}{\tau_*} A, qquad P(t_{\text{phys}}) = \exp\left[-\frac{t_{\text{phys}}}{\tau_*} A\right].$$

It also gives

$$c_* = \frac{\ell_*}{\tau_*}, \quad E_* = \frac{\hbar_*}{\tau_*}, \quad p_* = \frac{\hbar_*}{\ell_*}, \quad m_* = \frac{\hbar_* \tau_*}{\ell_*^2}, \quad E_* = m_* c_*^2.$$

These are dimensional identities, not derivations of observed constants, Lorentz invariance or particle mass.

The free action of $(\mathbb{R}_+)^3$ on the scale triple leaves all dimensionless $W0$ relations invariant. Thus CA installs units without choosing values. It is selector-blind and state-category-blind. Moreover, k_B or an equivalent temperature/entropy conversion and a bath/process law remain necessary before Shannon entropy becomes thermodynamic entropy.

4 Round III: ST1212–ST1226 — sector package SA

Let

$$X = \mathbb{Z}_{12} \times \{\pm 1\}$$

encode an origin and orientation/polarity. The dihedral group D_{12} of order 24 acts by rotations $(r, \lambda) \mapsto (r + k, \lambda)$ and reflections $(r, \lambda) \mapsto (k - r, -\lambda)$. Direct enumeration proves that the action on X is free and transitive. Thus X is a D_{12} torsor: symmetry alone cannot choose a point.

The minimal conditional sector package is

$$SA = (A_{\text{origin}}, A_{\lambda}, A_{\text{coupling}}).$$

Origin alone leaves two polarities; polarity alone leaves twelve origins; together they select one labelled frame. Without a coupling axiom the label has no dynamical effect.

The finite selector potential

$$V_{\text{ax}}(r, \lambda) = \mu \left[w_o d_{\mathbb{Z}_{12}}(r, r_0)^2 + w_{\lambda} \frac{1 - \lambda \lambda_0}{2} \right]$$

has a unique minimum at (r_0, λ_0) for every audited positive $w_o, w_{\lambda} \in \{1, 2, 3\}$. Its weights and scale are assumed. Conversely, the unique invariant probability measure on the transitive torsor is uniform with mass $1/24$ per branch and is selector-blind.

CA and SA solve independent group actions:

$$(\mathbb{R}_+)^3 \times X.$$

Neither supplies the other, and neither selects classical versus quantum states.

5 Round IV: ST1227–ST1241 — two-package insufficiency

Fix one identical data set

$$W0, qquad CA = (\ell_*, \tau_*, \hbar_*), qquad SA = (r_0, \lambda_0, \text{coupling}).$$

It admits at least the following normalized models:

1. classical probabilities on Δ_{11} with e^{-tA} ;
2. pure rays in $\mathbb{CP}^{11}$ with e^{-itA} ;
3. density states in $\mathcal{D}_{12}$ with unitary conjugation.

Their dimensions 11, 22, 143 and topologies remain inequivalent. CA only replaces t by t_{phys}/τ_* . SA can choose the same preparation label r_0 in every model.

For $r_0 = 0$ and dimensionless time $t = 1$, the return experiment gives

$$p_C(0|0; 1) = (e^{-A})_{00} = 0.2628294052832485,$$

$$p_Q(0|0;1) = |(e^{-iA})_{00}|^2 = 0.5843533247056469.$$

The difference is 0.3215239194223984. Hence the same $W0 + CA + SA$ supplies distinct testable predictions after a state/channel interpretation is chosen.

Theorem 1 (two-package insufficiency). *$W0 + CA + SA$ does not determine a unique operational theory. Units and a selected sector frame do not select the state category, physical channel, clock implementation, preparation, instrument or record.*

Proof. The three models above share identical $W0$, CA and SA , but their state spaces are pairwise inequivalent and the displayed common preparation/readout experiment has unequal predictions. Therefore those data cannot determine a unique operational model. $\square$

The documented adaptive form

$$\dot{K} = -\Pi(V'(K) - C(K))$$

does not repair the theorem. It is typed on an operator manifold and requires Π, V, C . The counterflow $\dot{A} = -\gamma A$ preserves both the Markov- Laplacian and self-adjoint quantum-generator interpretations. An adaptive law selects a category only if it uniquely preserves a category-specific cone or algebra and supplies an operational readout; no such strict theorem exists.

6 Round V: ST1242–ST1256 — operational package OA

The minimal interface found in this cycle is

$$OA = (\text{StateCategory}, \text{PhysicalChannel}, \text{ClockMap}, \text{Preparation}, \text{Instrument}, \text{RecordRule}).$$

A classical witness chooses Δ_{11} , heat evolution, the clock map $t = t_{\text{phys}}/\tau_*$, preparation δ_{r_0} , vertex readout and timestamped counts. A quantum witness chooses $\mathcal{D}_{12}$, unitary conjugation, the same dimensionless clock conversion, preparation $|r_0\rangle\langle r_0|$, a declared vertex PVM instrument and timestamped outcomes. They reproduce the distinct return probabilities above.

Each field is necessary in the declared operational class:

Removed field	Remaining obstruction
StateCategory	simplex, ray and density domains remain
PhysicalChannel	heat, unitary and a continuum of open channels remain
ClockMap	a unit exists, but no start/readout/calibration protocol
Preparation	a continuum of inputs gives different predictions
Instrument	one POVM admits different state-update maps and sequential statistics
RecordRule	no frozen event schema or falsifiable data product

Composition/environment is required for open dynamics and unbounded histories, but it is an extension rather than a minimal field for one finite experiment. No apparatus, raw event record or independent custody exists in the repository; the witnesses are executable specifications only.

7 Round VI: ST1257–ST1271 — three-package theorem

Theorem 2 (strategic three-package architecture). *Within the declared FIN conditional construction, the three obligation classes conversion, sector selection and operational meaning are independent. No two of CA , SA and OA cover all three. A conditionally dimensionful, sector-selected and testable theory therefore requires*

$$W0 + (CA \times SA \times OA) \longrightarrow W3_{\text{conditional}}.$$

Proof. Removing CA leaves every prediction dimensionless, and every proper subset of its three scale vectors has rank below three. Removing SA leaves the free transitive 24-frame torsor. Removing OA leaves the exact classical and quantum countermodels with unequal return predictions and no unique record semantics. The explicit product $(\mathbb{R}_+)^3 \times X \times \{OA \text{ choices}\}$ permits each factor to vary while the other two are fixed, proving independence. $\square$

The minimality is by obligation class, not by scalar count. A future theorem may derive several fields from one deeper object, but no such object is present.

8 Consequences for annihilation and dual dynamics

The one-nadsoliton ontology still distinguishes literal nonexistence from internal pattern loss. The package analysis now assigns responsibilities:

Layer	Contribution to the annihilation question
$W0$	provides candidate conserved mass, norm or trace in distinct channels
CA	gives those laws dimensional time and units, but adds no persistence
SA	labels an oriented branch, but does not define a survival observable
OA	selects the state/channel and specifies what is prepared, measured and recorded
ontology	excludes literal nonexistence only conditionally through complete state-space closure

Thus a unique operational statement about “what blocks annihilation” requires OA . In a classical OA it is mass conservation; in a quantum OA it is norm/trace conservation. Literal annihilation of the entire being still cannot leave an internal post-event record.

The dual dynamics remains a valuable mathematical property of $W0$:

$$e^{-itA}, \quad e^{-tA}.$$

It is not a physical choice until OA selects one channel or supplies a derived relation between them.

9 Updated strategic verdict

The previous two-package proposal was not wrong; it was incomplete. CA and SA remain the correct conditional answers to independent unit and sector cuts. Release 10.96 shows that neither addresses the state/operation cut.

Layer	Role	Status	Principal missing source
$W0$	informational mathematics	strict in declared scope	physical interpretation
CA	length/time/action conversion	conditional	scale-charged strict values
SA	origin/polarity/coupling	conditional	non-premise selector source
OA	state/channel/measurement/record	conditional	unique strict operational source or evidence
$W3$	effective physical predictions	conditioned on all packages	empirical validation and full dynamics

No conditional closure is a strict ToE result.

10 Recommended next programmes

Only nonreplay routes remain admissible:

1. publish $W0$ as standalone informational mathematics with its no-go map;
2. formalize typed interfaces and dependency checks for CA , SA and OA ;
3. freeze one explicitly conditional OA candidate rather than mixing channels;
4. produce a complete classical-versus-quantum prediction table for the frozen candidate;
5. prove global well-posedness for one specified nonlinear completion;
6. construct a refinement-limit internal environment, not a finite all-time heat dilation;
7. seek a genuinely new state-category source outside $C^*(A)$;
8. test a new S_+ only if a scale-charged formula is supplied;
9. test a new origin source only if it breaks translation by formula;
10. formalize package independence in a proof assistant;
11. design independent-record laboratory transfer only after OA is frozen;
12. audit legacy physical roles only after the legacy-to-strict bridge is completed.

Do not repeat functions of A as ontology selectors, weight-zero unit monomials, invariant frame measures, CA as a selector, SA as a unit source, adaptive terminology without a typed law, or receiver fits as sources.

11 Final verdict

The rigorous conditional FIN architecture has three, not two, independent missing bridges. Conversion axioms can supply units. Sector axioms can choose an oriented branch. Neither determines what counts as a state, which dynamics is physical, how time is read, how the state is prepared, what instrument acts, or what record falsifies the prediction. The operational package OA is mathematically unavoidable after the A -only nonuniqueness theorem. Current strict FIN remains $W0$; CA , SA , OA and every resulting $W3$ statement are conditional.

This is an important refinement of the strategic verdict, not physical confirmation of FIN.

A Programme ledger

Range	Main object	Verdict
ST1182–ST1196	$W0$ and operator algebras	no new strict source; $\mathcal{D} + A$ generates M_{12} conditionally
ST1197–ST1211	conversion package CA	rank-three dimensional closure, free scale torsor
ST1212–ST1226	sector package SA	exact frame torsor and conditional selection
ST1227–ST1241	two-package falsification	same CA/SA , inequivalent predictions
ST1242–ST1256	operational package OA	six necessary interface fields

Range	Main object	Verdict
ST1257–ST1271	strategic theorem	three independent packages required

B Reproducibility

The authoritative result sets run from `FIN_ST1182_ST1196_Results.json` through `FIN_ST1257_ST1271_Results.json`. Each contains fifteen programmes and SHA-256 references to individual packets. The combined test `test_fin_st1182_st1271.py` checks all ninety hashes, the algebra dimensions, *CA/SA* minimality, the two-package countermodel, *OA* status and the final theorem scope.

C Nonclaims

No new strict unit, selector, state category, adaptive source law, physical clock, apparatus, raw event, independent audit, legacy bridge/role transfer, Lorentz theory, Standard Model, gravity, L_{total} or Theory of Everything is established. All future FIN PDFs remain English.

References

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